

PATIENT HISTORY:

CHIEF COMPLAINT/PRE-OP/POST-OP DIAGNOSIS: Thyroid nodule
PROCEDURE: Total thyroidectomy
SPECIFIC CLINICAL QUESTION: Not answered
OUTSIDE TISSUE DIAGNOSIS: Not answered
PRIOR MALIGNANCY: Not answered
CHEMO RADIATION: Not answered
ORGAN TRANSPLANT: Not answered
IMMUNOSUPPRESSION: Not answered
OTHER DISEASES: Not answered

UUID: 963F72D4-5416-48E1-972B-2DD8ED0868E8
TCGA-BJ-A0ZC-01A-PR

Redacted



1CD-0-3

carcinoma, papillary, follicular variant

8340/3

Site: Thyroid, NOS C73.9

lw
11/27/11

ADDENDA:**Addendum****MOLECULAR ANATOMIC PATHOLOGY TESTING:****Block I:**

- A. Mutations in **BRAF**, **NRAS61**, **HRAS61**, **KRAS12/13** NOT identified.
B. FISH test results are negative for **RET/PTC** rearrangement.

NOTE:

DNA was extracted in the amount sufficient for testing.

BACKGROUND:

Mutations in either **BRAF** or **RAS** genes or **RET/PTC** rearrangements are found in more than 70% of papillary thyroid carcinomas (1). **BRAF** V600E (T1799A) mutation has been associated with more aggressive behavior of papillary carcinoma (2, 3). The association between **BRAF** V600E mutation and features of tumor aggressiveness have also been observed in papillary microcarcinomas (4). Mutations in the **RAS** genes or **PAX8/PPARg** rearrangement occur in ~70% of follicular thyroid carcinomas and with lower frequency in oncocyctic (Hürthle cell) carcinomas (5). Regarding the specificity of these mutations for cancer, **BRAF** V600E mutation and **RET/PTC** and **PAX8/PPARg** rearrangements are overall specific for malignancy in the thyroid, although they have been reported with a very low frequency in benign thyroid lesions (6). **RAS** mutations occur in malignant and benign thyroid tumors, being found in ~40-50% of follicular and anaplastic carcinomas, 30-40% of follicular adenomas and 10-15% of papillary carcinomas (6).

- Adeniran AJ, et al. Correlation between genetic alterations and microscopic features, clinical manifestations, and prognostic characteristics of thyroid papillary carcinomas. *Am J Surg Pathol* 2006, 30:216-222
- Xing M, et al. **BRAF** mutation predicts a poorer clinical prognosis for papillary thyroid cancer. *J Clin Endocrinol Metab* 2005, 90:6373-6379.
- Elisei R, et al. **BRAF** V600E mutation and outcome of patients with papillary thyroid carcinoma: a 15-year median follow-up study. *J Clin Endocrinol Metab* 2008, 93:3943-3949.
- Lee X, Gao M, Ji Y, Yu Y, Feng Y, Li Y, Zhang Y, Cheng W, Zhao W. Analysis of differential **BRAF**(V600E) mutational status in high aggressive papillary thyroid microcarcinoma. *Ann Surg Oncol*. 2009 Feb;16(2):240-5.
- Nikiforova MN, et al. **RAS** point mutations and **PAX8-PPARγ** rearrangement in thyroid tumors: Evidence for distinct molecular pathways in thyroid follicular carcinoma. *J Clin Endocrinol Metab* 2003, 88: 2318-26.
- Nikiforov YE. Recent developments in the molecular biology of the thyroid. In: Lloyd RV., ed. *Endocrine Pathology: Differential Diagnosis and Molecular Advances*. Humana Press, Totowa, 2004, 191-209.

MUTATIONAL ANALYSIS:

For paraffin-embedded surgical specimens, manual microdissection was performed to collect tumor tissue. Specimens with the minimum of 50% of tumor cells in a microdissection target are accepted for the analysis. Optical density readings were obtained. Real-time PCR was performed on the platform to amplify **BRAF** codons 599-601, **NRAS** codon 61, **HRAS** codon 61, and **KRAS** codons 12/13 sequences. Post-PCR melting curve analysis was used to detect possible mutations. If required, the mutation type was confirmed by Sanger sequencing of the PCR product on DNA from samples positive for each of these mutations was used as positive controls. Amplification at 35 cycles or earlier was considered sufficient for the analysis. The limit of detection is approximately 10-20% of alleles with mutation present in the background of normal DNA.

FINAL DIAGNOSIS:

THYROID GLAND, TOTAL THYROIDECTOMY (29.5 GRAMS) -

- PAPILLARY MICROCARCINOMA, FOLLICULAR VARIANT, 0.4 CM, LEFT LOBE (see comment).
- NODULAR THYROID (LARGEST NODULE 3.5 CM, SOME OF WHICH SHOW FIBROSIS AND OSSIFICATION).
- CHRONIC LYMPHOCYTIC THYROIDITIS.
- NO PARATHYROIDS ARE SEEN.
- STAGE: pT1 NX MX.

COMMENT:

We have demonstrated a focus of papillary microcarcinoma, follicular variant (0.4 cm) in the left lobe of the thyroid. The margins are free of tumor. No extrathyroidal extension or angiolymphatic invasion is seen.

Criteria	Yes	No
Diagnosis Discrepancy		
Primary Tumor Site Discrepancy		
HIPAA Discrepancy		
Prior Malignancy History		
Dual/Synchronous Primary Notes		
Case is (Circle):	QUALIFIED	DISQUALIFIED
Reviewer Initials	lw	
Date Reviewed	11/27/11	

CASE SYNOPSIS:

SYNOPTIC DATA - PRIMARY THYROID TUMORS

SPECIMEN TYPE: Total Thyroidectomy
TUMOR SITE: Left Lobe ✓
TUMOR FOCALITY: Unifocal ✓
TUMOR SIZE (largest nodule): Greatest Dimension: 0.4 cm
Additional Dimensions: 0.4X0.4 cm
HISTOLOGIC TYPE: Papillary carcinoma ✓
PATHOLOGIC STAGING (pTNM): pT1
pNX
pMX
MARGINS: Margins uninvolved by carcinoma
VENOUS/LYMPHATIC (LARGE/SMALL VESSEL) INVASION (V/L): Absent
ADDITIONAL PATHOLOGIC FINDINGS: Nodular goiter
Thyroiditis

SPECIAL PROCEDURES:

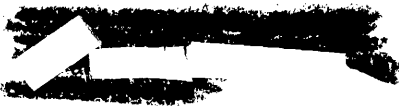
In Situ Procedure

Interpretation

RESULTS:

1i: *RET/PTC*: The targeted area of the tissue showed 3(5.0%) of the cells with the rearrangement pattern and 57(95.0%) of the cells with the normal pattern. The targeted area is considered NOT rearranged for the *RET/PTC* region.

The *RET/PTC1* Probe did not contain the rearrangement pattern.



My signature is attestation that I have personally reviewed the submitted material(s) and the above diagnosis reflects that evaluation.

Results

RET/PTC Methodology Description:

Probe: *RET/PTC* Probe (10q11.2) and *RET/PTC1* Dual-Color Fusion Probe 10q11.2-10q21

Probe Description: The *RET* probe (labeled SpectrumGreen) is a 207 kb DNA probe located at band 10q11.2.

The H4 and 369 probes are approximately 100kb each (labeled in SpectrumOrange), and are located at band 10q21.

RET/PTC rearrangement is suggested to be present when three *RET* signals are identified in greater than 8% of the analyzed cells.

RET/PTC1 rearrangement detection is as follows: Normal cells without the *RET/PTC1* rearrangement show two pair of green and orange signals separated. Cells with the *RET/PTC1* rearrangement show two pair of signals (green and orange) in juxtaposition and one green and one orange signal separated in greater than 8% of analyzed cells within the targeted area.

RET/PTC FISH analysis was manually performed and quantitatively assessed by analysis of a minimum of 60 cells using the *PTC* (369/H4) and the *RET* SpectrumGreen probes.